

SMART UNIVERSITY ATTENDANCE TRACKING SYSTEM PROJECT PROPOSAL (U-PANEL)

Prepared for: Kampala International University (KIU)

Prepared by: Development Team

Date: September 2026

Document: Project Proposal

Total project cost: UGX 150,000,000 (exclusive of applicable taxes /VAT)

Proposal validity: 60 days from the date above

Document type: Commercial and Technical Proposal

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Executive Summary

U-Panel is an attendance system for Kampala International University (KIU). A lecturer opens a live session; students mark on their phones; administrators pull reports. The product is meant to sit beside KIU's existing academic and student systems, not replace them.

Student master data already lives in KIU systems. U-Panel will call the APIs KIU provides for that data. It will also publish its own APIs so other KIU systems can read attendance results and plug into workflows.

We store sessions, submissions, attendance results, and the only references or caches needed to run securely. We don't become a second student registry.

The application runs on infrastructure KIU controls (on prem or cloud under KIU). KIU owns all institutional data. After full payments and acceptance, KIU gets the complete source code and ownership of the custom U-Panel application code built under this engagement. Third-Party and open-source components keep their own licenses.

Background & Vision

1.1 Background

Paper registers, sign-in sheets and spreadsheets are still common. They slow departments down, create entry errors, and leave academic leadership without a clear real-time picture across faculties and campuses.

1.2 Vision

Build an attendance layer that plugs into what KIU already runs. Students, lecturers, departments, and administrators share one live record without standing up another student master database.

Objectives & Solution Overview

1.3 Core objectives

1. More off-paper attendance and reduce manual consolidation
2. Let authorized lecturers start and close live attendance sessions
3. Give students a simple, secure way to submit attendance
4. Improve accuracy of records and institutional reporting
5. Support use across KIU campuses as infrastructure allows
6. Integrate both ways: Consume KIU APIs for student academic data, and publish U-Panel APIs so attendance sits in the institutional stack rather than as a silo.
7. Improve the time-frame for report generation for all departments and directories

1.4 User Groups

- **Students:** sign-in securely; join an active session; submit attendance; view personal history
- **Lecturers:** create sessions; generate session codes/tokens; watch submissions live; review logs; close sessions manually or on a timer.
- **University Administrators:** manage users, roles, permissions, academic structures, and courses; run institution-wide reports.
- **ICT/ System Administrators:** deploy and operate services; manage environments; apply updates from source; handle backups and access control under KIU IT policy.

Extra roles (Head of Department, Registrar view-only, and similar) can sit under RBAC within the agreed customization. We confirm these in requirements.

Platform Components and Technical Architecture

1.5 Core Components

- **Student Attendance** – Authentication, session matching, code/ credential submission, validation and confirmation.
- **Lecturer Management** – session creation, dynamic token/code generation live tracking, manual or timed closure.
- **Attendance administration** – role permissions and attendance settings, Student and course master data come from KIU’s existing systems via API, not from a separate standalone student DB.
- **Records & Reporting** – filter by student, lecturer, course, class, department, date, or academic period. Export as **CSV** and **PDF**.

1.6 Technical Stack

Layer	Technology
Mobile Client	Flutter – Android and iOS (student, lecturer and QA monitor mobile apps in scope).
Web administrative interface	Django admin / Django templates (server- rendered admin with the Django Backend)
Backend	Django / Python, REST APIs.
Database	PostgreSQL
Auth & APIs	Token/session-based API auth over HTTPS
Push Notifications	Built into the app when configured. Default: OneSignal (Android + iOS). Other providers possible if KIU standardizes elsewhere. Account, API keys, and billing sit with KIU.
Email	Built into the app when configured. Default: Mailjet . Other SMTP/API providers possible. Account, keys and billing sit with KIU.
CDN	KIU-provided CDN (or equivalent) for app assets and static content with account and billing under KIU.

1.7 Logical Architecture

Flutter apps (students and lecturers) and the web admin call the **U-Panel Django Rest API** over HTTPS. The API reads and writes attendance data in U-Panel’s PostgreSQL store. For student and academic master data it calls KIU’s existing-system APIs.

Other KIU systems can call **U-Panel Integration APIs** for attendance results, session status and report hooks as agreed.

RBAC and audit logging sit in the application layer. Everything runs inside KIU’s network and security boundary. **U-Panel is not a standalone SIS.**

1.8 Integration model

KIU to U-Panel: KIU provides documented, authenticated APIs (and a sandbox where available); for-example student identity/ enrolment, courses/classes, lecturer assignments, and directory hooks if applicable.

U-Panel to KIU systems: The Development Team builds and documents U-Panel APIs for customers; attendance marked events, session open/closed, per-course summaries, and export hooks.

What is needed for integration

- Stable API contracts, credentials and environments (sandbox and production)
- A named API owner on the KIU side for schema questions and change notice
- Agreement during requirements on sync style (real-time calls and/or scheduled sync)

What we are not doing: replacing KIU's student information system, or becoming the system of record for student biodata.

Infrastructure Requirements, Hosting & Data Ownership

U-Panel runs entirely on **KIU-controlled infrastructure**. KIU provisions and funds what is listed below. The Development Team deploys and configures the application against it.

1.9 Environments

- **Staging** – User Acceptance Testing, training, pre-release checks.
- **Production** – live institutional use.

Staging and production should be separate virtual machines/hosts (or separate cloud projects). Keep the same OS and major software versions on both.

1.10 Server sizing (hardware requirements)

Every app or database sever needs at least 8GB RAM. Do not deploy on smaller hosts.

Concurrent attendance sessions are **not** capped in software; throughput follows this hardware.

Role	Staging (minimum)	Production (minimum)
Application server (Django + reverse proxy)	2 vCPU – 8 GB RAM – 40 GB SSD	4 vCPU – 8 GB RAM (16 GB preferred at peak) -80 GB SSD
Database server (PostgreSQL 14+ /16)	Dedicated preferred; hosts still >= 8 GB RAM	Dedicated: 2-4 vCPU – 8 GB RAM (16 GB preferred) – 100 GB +SSD
File / media storage	Local disk- acceptable initially	Dedicated volume or S3 - compatible bucket

OS: Ubuntu LTS 22.04 or 24.04 (or KIU's standard equivalent Linux).

1.11 Platform software

- **App runtime:** Python 3.11+ (version pinned at deploy).
- **Web / TLS:** Nginx (or Caddy); valid HTTPS certificate.
- **Database:** PostgreSQL (not exposed to the public internet).
- **Process control:** system (or equivalent).
- **Time sync:** NTP /chrony (accurate timestamps).
- **Admin UI:** Django admin/ Django templates.

1.12 Network & DNS

- **Public URL:** HTTPS hostname chosen by KIU (e.g. upanel.kiu.ac.ug)
- **DNS:** points to the reverse proxy
- **Firewall:** 443/tcp public; SSH/admin only via VPN or bastion; allow 80/tcp only if needed for certificate renewal.
- **Client access:** mobile apps must reach the API reliably on- and off-campus.
- **Outbound HTTPS:** app servers must reach OneSignal (or agreed push), Mailjet (or agreed email) and CDN endpoints.

1.13 Messaging & edge services (configured in the app)

Email and push are built into U-Panel. They switch on once provider credentials are in place. Recurring provider fees and accounts remain with KIU.

- **Push Notifications** – default OneSignal. KIU provides the OneSignal app ID /API key (and store-related setup as required). Used for session alert, reminder and mobile notices.
- **Email** – default Mailjet. KIU provides Mailjet API key/ secret (or SMTP credentials) and verified sending domain (SPF/DKIM). Used for OTPs and admin/lecturer/student mail.
- **CDN** – infrastructure side. KIU provides Cloudflare, CloudFront, Azure CDN, or KIU'S existing CDN for static/assets delivery and edge caching. If KIU already standardizes on another email or push vendor, we can agree and alternate in requirements. Those channels still need valid configuration before they send.

Credentials to share securely with the team: Mailjet (or alternate email) keys and from domain; OneSignal (or alternate push) keys; CDN distribution URL or pull zone.

Late credentials do not block core attendance UAT (open session>> submit>>report). Email and push stay off until configured.

1.14 Backups (KIU-owned)

- **Database:** daily automated backups.
- **Offsite:** at least one copy off the primary host.
- **Restore test:** successful restore to staging once before go-live. RPO/RTO follow KIU ICT policy.

1.15 Temporary access for the Development Team

Until handover we need deploy access to staging (and production during go-live windows), rights to install packages and run migrations, a DNS/TLS contact, and a named ICT counterpart. Access is revoked or reduced at handover as KIU prefers.

1.16 Infrastructure-ready checklist

- Staging and production hosts live (**>= 8GB RAM** for app and DB roles)
- PostgreSQL installed (production dedicated preferred)
- Domain + TLS on HTTPS
- Firewall rules applied
- **OneSignal** (or agreed push) credentials issued
- **Mailjet** (or agreed email) credentials and from-domain ready
- CDN configured (or interim origin-only agreed in writing)
- Backups configured; restore tested once
- Deploy access issued to the Development Team
- ICT contact named for go-live week

1.17 Data ownership

KIU retains **absolute ownership** of all institutional data. Student master records and related academic structures stay owned and authoritative in KIU's existing systems. U-Panel runs the attendance store (sessions, submissions, results, audit logs) on KIU infrastructure. Retention and disaster-recovery policy remain KIU's.

1.18 Not required for v1

Separate Windows servers; Oracle/MSSQL; SMS/ WhatsApp gateway hardware; on-prem Apple or Google build machines (the Development Team produces mobile builds)

Security, Attendance Integrity, Source Code & Rights

1.19 Application security

- Secure user authentication
- Role-based access control (RBAC)
- Encrypted API traffic (TLS)
- Server-side validation of every attendance submission
- Session management with expiry
- Application audit trails for sensitive actions

KIU secures the network, hosts and physical or cloud environment under institutional policy.

1.20 Attendance integrity controls (in scope)

To reduce proxy marking and casual abuse, U-Panel includes these **application-level** controls. Exact parameters are set during requirements;

- **Time-bounded sessions** – attendance only while the lecturer’s session is open
- **Dynamic/ rotating sessions or tokens** – codes are not static for the whole term
- **One valid submission per student per session** -blocks duplicate marks
- **Lecturer visibility of live sessions** -spot anomalies during the session
- **Audit log of submissions** (timestamp, student, session)- supports later review

Not in base-scope: biometric hardware, facial recognition, or invigilation-grade exam proctoring (*see Out of Scope*). Those can be a later phase if KIU wants them.

1.21 Data protection

The system will support KIU’s obligations under applicable law, including Uganda’s **Data Protection Privacy Act**, by providing:

- Role limited access to personal data
- Authentication and audibility of access
- Ability for KIU to apply retention and depletion policies at the infrastructure/data layer
- KIU remains the **data controller** for institutional personal data. The Development Team only processes configuration and support access as needed to deliver the project, under KIU direction.

1.22 Intellectual property. Source code & usage rights

Upon full payment and written acceptance:

- **Custom U-Panel application code** developed under this project (mobile app source, backend source, admin UI source, project-specific scripts and configs authored by the team) is **assigned to KIU**. (KIU campuses, branches, and affiliated entities.)
- KIU receives a complete, documented source-code handover (repository access or archive as agreed).
- **Third -party and open-source components** remain under their respective licenses. The team will provide a bill of materials/ license list at handover,
- **Pre-existing tools, libraries, or know-how** owned by the Development Team (if they are reused) are licensed to KIU on a perpetual, royalty-free basis for use within U-Panel at KIU; *to be listed if applicable*.

Restrictions for the system

- Deployment and use under this fee are for the **KIU institutional group only**.
- Sublicensing, resale or transfer to unrelated external organizations needs a **separate written agreement**.

If procurement prefers “perpetual license +escrow” instead of copyright assignment, that clause can be swapped in the contract without changing the technical delivery.

Scope (In / Out), Assumptions & Dependencies

1.23 In scope

1. Requirements confirmation (workflows, roles, reporting, **integration contracts**)
2. Integration with KIU; provided APIs for student and academic **master data** (and related lookups agreed in requirements).
3. Design and delivery of **U-Panel APIs** for consumption by KIU's existing systems.
4. Configuration of roles/permissions and attendance settings (master lists driven by KIU systems where applicable).
5. Agreed KIU-specific customization within the timeline envelope.
6. Backend, database, **Android and iOS** mobile apps, and Django admin deployment onto KIU environments.

7. Structured testing (*including integration tests*), UAT support, training, documentation, and production handover.
8. Attendance integrity controls listed in *Sec. 1.20*.
9. **Email and push notification features** in the application, enabled when configured (defaults: **Mailjet, OneSignal**).
10. No artificial software/license cap on concurrent live attendance sessions (capacity follows KIU infrastructure).
11. Staging + production setup guidance and go-live support during stabilization.

1.24 Out of scope (unless added by change request)

- Replacing or re-platforming KIU's existing student information / academic core systems
- Building KIU's upstream APIs (KIU provides and maintains those interfaces)
- Biometric devices, turnstiles, or hardware procurement
- Facial recognition or continuous exam proctoring
- SMS/WhatsApp gateway fees; **Mailjet / OneSignal (or alternate) and CDN accounts and usage fees** (KIU-funded)
- Multi-tenant SaaS for other universities
- Ongoing App Store / Play Store **renewal** fees after handover if listings remain on Development Team accounts (Year-1 store fees during launch are covered by the team when using team accounts; KIU may later transfer listings to its own accounts)
- Ongoing production hosting, bandwidth, SSL renewal, and 24/7 operations after handover
- Post-warranty feature development (available under separate support/CR)
- Historical paper-register digitization
- Custom data-warehouse or BI tool licensing

1.25 Assumptions

- KIU nominates a single focal person / product owner who can decide on requirements within 3 business days.
- KIU provides production and staging infrastructure matching the published requirements
- U-Panel does not operate as a standalone student database; identity and enrolment truth remain in KIU's existing systems.
- Roll-out model: academic year by year. Year 1 first until complete; later years onboard after Year 1 (schedule agreed with KIU).
- Concurrent live sessions: no artificial license / product cap – unlimited concurrent sessions in software; peak throughput follows KIU infrastructure capacity.
- Mobile apps ship for both Android and iOS at go-live.
- English is the primary UI language for v 1.

Note on concurrent live sessions: *several lecturer sessions may be open at once. Software does not limit how many can run. Peak load follows KIU's hosting capacity.*

1.26 Dependencies

- **KIU server/ cloud readiness** - delay slides the timeline
- **KIU existing system APIs** ready (docs, credentials, sandbox) -delay blocks integration and student lookup.
- **DNS, SSL, Firewall rules** – delay blocks production URL and mobile API access.
- **OneSignal (or agreed push) credentials** -push inactive until configured.
- **Mailjet (or agreed email) credentials** -email inactive until configured.
- **CDN endpoint** (if static assets are CDN-fronted) – affects asset performance; app can fall back to origin temporarily.
- **Named UAT participants** from academic + ICT delay – blocks the acceptance milestone.
- **Apple / Google developer account availability** - affects whose store listing is used; not a blocker to shipping both platforms

Testing, Acceptance Criteria, Training & Deployment

1.27 Testing & QA

We cover functional workflows, student submission, lecturer controls, authentication, permissions, data validation, basic performance under expected concurrent overload, and application security checks. KIU runs **User Acceptance Testing (UAT)** on staging.

1.28 Acceptance Criteria (gates final 20% payment)

Acceptance holds when **all** of the following are true (or waived in writing by KIU):

1. Agreed in-scope modules are deployed to **KIU production** and reachable over HTTPS.
2. Staging and production configuration are documented; source code is delivered as agreed.
3. The jointly agreed UAT script is complete with no **Criteria** or **High** open defects (severity definitions in the test plan). Medium/Low items may stay on a punch-list with owners and dates.
4. Admin, lecturer, and ICT training sessions are delivered (or recorded materials provided if KIU reschedules).
5. Handover pack delivered: credentials process, runbooks, API/ admin overview, license BOM, test summary.
6. KIU signs a short **Acceptance Certificate** (or email equivalent from the Project Sponsor).

1.29 Training & knowledge transfer

- **ICT/ System Administrators** – deploy, backup restore basics, logs, user provisioning, update process.
- **University Administrators** - Academic structure, roles, reports.
- **Lecturers**- session create/ monitor/close; common student issues.

1.30 Production deployment

Stabilized build goes to KIU production. Administrators are set up. Smoke tests run. Then formal handover into the warranty period.

1.31 Mobile distribution (android & iOS)

Apps ship on **Google Play (Android)** and the **Apple APP (I OS)**.

- **Preferred:** Publish under **KIU's** Apple Developer and Google Play accounts when KIU has (or creates) them. Clearest ownership of the store listings.
- **Fallback:** if those accounts are not ready at launch, the Development Team publishes under **the team's developer accounts** so staff and students can install from the official stores without waiting.

- When KIU later gets institutional accounts, the team will support **transfer of the app listings** to KIU (subject to Apple/Google rules at that time).

Store listing content (name, screenshots, privacy text) uses KIU branding as agreed. Side-load/ MDM remains a contingency if store review blocks go-live timing

Support, Warranty, Change Control, Deliverables & Governance

1.32 Warranty (included)

For **six (6) months** after written acceptance, the Development Team will fix **defects** in delivered in-scope functionality (failure to perform as specified) at no extra charge, Monday-Friday, **09:00-17:00EAT**.

Warranty **excludes**: new features, KIU infrastructure failures, third-party outages, unauthorized modifications, and issues from data or config errors introduced by KIU after handover.

1.33 Post-handover support (optional, not in base fee)

Ongoing maintenance, enhancements, and SLA-based support are available under a **separate support agreement** (monthly/annual retainer or time-and-materials). Quote on request.

1.34 Change control

Scope changes after requirements sign-off go through a written Change Request (CR): description, cost/timeline impact, and KIU approval before work starts. CRs are billed separately unless both sides agree in writing to absorb them.

1.35 Deliverables

- Fully deployed U-Panel (mobile + admin + backend APIs + DB configuration)
- Institutional settings as configured
- Complete source-code access/ archive
- Technical documentation and operational runbooks.
- **Integration API documentation** (U-Panel endpoints + expected KIU API usage)
- License / third-party BOM
- Testing summary and UAT records
- Acceptance Certificate

1.36 Governance

Named KIU representatives (as KIU assigns) work with the Development Team Project Lead and Technical Lead. Weekly status (or as agreed). Escalate blockers within 2 business days.

Responsibilities

1.37 Kampala International University (KIU)

- Provide production and staging infrastructure, network, DNS, SSL, security resources, **CDN**, and credentials for **email (default Mail jet) and push (default OneSignal)** so those app features can be enabled.
- Provide and maintain **APIs into existing systems** (student database and related academic data), including documentation, credentials, sandbox, and a named API owner.
- Nominate a product owner / focal person and UAT participants.
- Participate in requirements confirmation, UAT, and timely approvals.
- Manage production operations, backups, retention, and post-handover administration.
- Fund infrastructure and third-party operational costs excluded from the project fee

1.38 Project Team

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- Design, build, configure, customize, and deploy U-Panel per agreed scope.
- Integrate against KIU-provided API and deliver **U-Panel APIs** for KIU system consumers, with documentation.
- Provide technical requirements, documentation, and agreed training.
- Execute structured testing and support UAT.
- Complete source handover and production acceptance support.
- Honor the **6-month** warranty period for in-scope defects.

Commercial Proposal & Payment Terms

1.39 Project fee

Total project cost: UGX 150,000,000 (*One hundred fifty million Uganda Shillings*)

- **Tax treatment:** project fee is **exclusive of applicable taxes/VAT**; tax charged separately on invoice if legally required.
- **Currency:** UGX
- **Payment method:** bank transfer to Development Team account on invoice. **Bank/account details will be provided after contract signing.**

STAGE	WHAT THE COST COVERS	AMOUNT
Personnel & Technical Development	Software engineers, consultations, backend development, mobile development, frontend, database and technical personnel consultations <i>for example senior lecturers,</i>	UGX 58,000,000
Requirements analysis, UI/UX & Product Design	Requirements analysis, system architecture, workflows, UI/UX and prototypes	UGX 12,000,000
Project Management & Coordination	Planning, coordination, meetings, reporting, documentation and project supervision	UGX 15,000,000
System integration & APIs	Integration with internal application APIs, <i>U-Panel APIs and other required services</i>	UGX 18,000,000
Infrastructure, DevOps & Deployment	Development/staging environments, production deployment, configuration, hosting-related setup and handover	UGX 18,000,000
Quality Assurance, Testing & UAT	Functional Testing, security testing, performance testing, bug fixing and User Acceptance Testing	UGX 12,000,000

Operational & Communication Costs	Project communication, stakeholder engagement, meetings, coordination and related operational expenses	UGX 5,000,000
Training, Documentation & Knowledge Transfer	Staff training, user manuals, technical documentation and handover	UGX 5,000,000
Contingency- Scope & Schedule Risk Buffer	Provision for unforeseen technical and implementation requirements within the agreed scope	UGX 2,000,000
Administrative & Business Overhead	Administrative, legal, financial, software/tools and general business operating costs associated with delivery	UGX 13,000,000
TOTAL PROJECT FEE		UGX 150,000,000

RETURN ON INVESTMENT (ROI) PROJECTION

The proposed system requires an initial investment of UGX. 150,000,000. This investment is expected to generate financial and operational benefits through automation, improved revenue collection, reduced paperwork, and improved accountability.

1. ESTIMATED ANNUAL BENEFITS

Benefit Area	Estimated Annual Value
Reduction in Administrative and manual processing costs	UGX 25,000,000
Reduction in printing, stationery and physical record costs	UGX 10,000,000
Staff time saved through automation	UGX 20,000,000
Improved revenue collection and reduced revenue leakage	UGX 30,000,000
Reduction in errors and associated correction costs	UGX 10,000,000
Improved operational efficiency and reporting	UGX 15,000,000
Total Estimated Annual Benefit	UGX 110,000,000

2. INVESTMENT PAYBACK

Initial System Investment: UGX 150,000,000

Estimated Annual Benefit: UGX 110,000,000

Estimated Payback Period: 1.4 years. Therefore, the university is projected to recover the initial investment in approximately **16 months**, subject to actual system usage and realized savings.

3. 3-YEAR ROI PROJECTION

YEAR	ESTIMATED BENEFIT	CUMULATIVE BENEFIT	NET POSITION
Year 1	UGX 110,000,000	UGX 110,000,000	-UGX 40,000,000
Year 2	UGX 110,000,000	UGX 220,000,000	+UGX 70,000,000
Year 3	UGX 110,000,000	UGX 330,000,000	+UGX 180,000,000

Over 3 years:

Total Benefits = UGX 330,000,000

Initial investment = UGX 150,000,000

Net Financial Benefit = UGX 180,000,000

4. ESTIMATED 3-YEAR ROI

The expected ROI for 3 years after rollout is calculated as:

ROI = (Total Benefits – Initial Investment) / Initial Investment x 100

Estimated 3-Year ROI = (330,000,000 – 150,000,000) / 150,000,000 x 100

Estimated 3-Year ROI = 120%

This means that based on the projected benefits, every **UGX 1 invested** in the system could generate approximately **UGX 2.20 in total benefits over 3 years**, including recovery of the original investment.

5. LONG-TERM VALUE

- Beyond the direct financial return, the university will also benefit from:
- Faster and more efficient service delivery
- Centralized and secure institutional information
- Improved management decision-making
- Better accountability and transparency
- Reduced dependence on manual processes.
- Improved tracking and reporting
- A scalable digital platform that can support future university needs

ROI SUMMARY

Initial Investment: UGX 150,000,000

Estimated Annual Benefit: UGX 110,000,000

Estimated Payback Period: ~16 Months

Projected 3-Year Benefits: UGX 33,000,000

Projected 3-Year Net Benefit: UGX 180,000,000

Projected 3-Year ROI: 120%

NOTE: The above figures represent a projected ROI based on estimated operational and functional benefits. Actual returns will depend on the university's transaction volumes, current administrative costs, revenue collection efficiency, adoption rate and level of system utilization.

1.40 Exclusions from the fee

Physical/cloud servers, internet bandwidth, third-party APIs, **OneSignal** (alternative push), **Mailjet** (or alternate email), **CDN** costs, SMS gateways, SSL by KIU.

1.41 Payment milestones

Milestone	Share	Amount (UGX)	Trigger
Contract signing & project commencement	60%	90,000,000	Signed contract + invoice
Completion of implementation & deployment (staging ready/ UAT start)	20%	30,000,000	Agreed build on KIU staging; UAT invited
Final testing, handover & acceptance	20%	30,000,000	Acceptance criteria in <i>Sec. 9.2</i> met

The above stated payment details are subject to change upon discussion with the KIU administration.

Amounts are exclusive of applicable taxes /VAT

Commercial and delivery snapshot: Project fee is UGX 150,000,000 (exclusive of applicable taxes/VAT). Indicative timeline is about 10-16 weeks, subject to infrastructure readiness, API access, and timely decisions. Rollout is year by year; Year 1 first until complete, then later academic years.

Risks (Summary)

- **Infrastructure not ready**- early requirements pack; staging-first; timeline contingency.
- **KIU APIs late, unstable, or undocumented**- API contract workshop early; sandbox gate before UAT; timeline contingency.
- **Slow UAT / approvals**- named KIU focal person;
- **Proxy attendance expectations exceed app-level controls.**
- **App store review delays**- parallel MDM/APK path; start store setup early.
- **Scope**- Change Request process (*Sec. 1.34*).

Conclusion

U-Panel is a straightforward way for KIU to run digital attendance without replacing what is already in place. Student master data stays in KIU's existing databases. Attendance runs on KIU-controlled hosts.

Other systems can consume attendance through APIs. Source code is handed over on acceptance, with a 6-month warranty on in-scope defects.

Ready to start once the contract is signed and commencement milestone is received.

Approval & Signature

For the Project Team

Name: _____

Title: _____

Signature: _____

Date: _____

For Kampala International University

Name: _____

Title: _____

Signature: _____

Date: _____

Appendix A – Glossary

UAT- User Acceptance Testing

CR- Change Request

EAT- East Africa Time

RBAC - Role-Based Access Control

BOM – Bill of Materials (third-party license)

LTS – Long Time Support

RTO- Recovery Time Objective

RPO- Recovery Point Objective